

Retail BI Platform

Data Warehouse Design & Implementation

Project: Olist E-commerce Business Intelligence Platform

Technology:

- PostgreSQL
- Talend Open Studio
- SQL
- Star Schema

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Phase: Data Warehouse

Date: August 2026

1. Introduction

This phase focuses on transforming the prepared Olist e-commerce data into a centralized analytical Data Warehouse.

The Data Warehouse was implemented using PostgreSQL and populated through ETL processes developed with Talend Open Studio.

The main objective is to provide a reliable and structured foundation for Business Intelligence analysis and reporting.

2. Data Warehouse Architecture

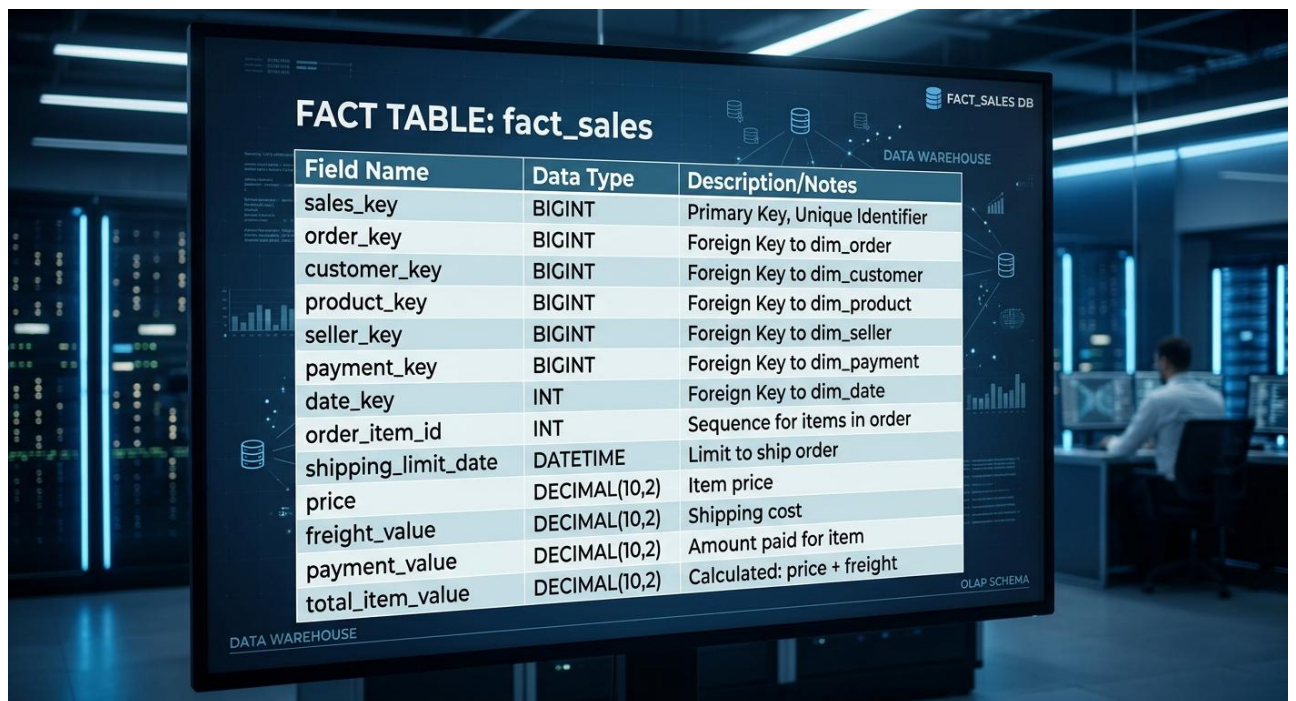


The staging area acts as an intermediate layer between the source data and the analytical Data Warehouse. Talend is used to transform, validate, and load the data into the dimensional model.

3. Dimensional Model

The staging area acts as an intermediate layer between the source data and the analytical Data Warehouse. Talend is used to transform, validate, and load the data into the dimensional model.

3.1 Fact Table and fields



Field Name	Data Type	Description/Notes
sales_key	BIGINT	Primary Key, Unique Identifier
order_key	BIGINT	Foreign Key to dim_order
customer_key	BIGINT	Foreign Key to dim_customer
product_key	BIGINT	Foreign Key to dim_product
seller_key	BIGINT	Foreign Key to dim_seller
payment_key	BIGINT	Foreign Key to dim_payment
date_key	INT	Foreign Key to dim_date
order_item_id	INT	Sequence for items in order
shipping_limit_date	DATETIME	Limit to ship order
price	DECIMAL(10,2)	Item price
freight_value	DECIMAL(10,2)	Shipping cost
payment_value	DECIMAL(10,2)	Amount paid for item
total_item_value	DECIMAL(10,2)	Calculated: price + freight

4. Dimension Tables

Dimension Table	Primary Key	Business Key	Main Attributes	Purpose
dim_customer	customer_key	customer_id	customer_unique_id, customer_zip_code_prefix, customer_city, customer_state	Stores customer information and geographic details
dim_product	product_key	product_id	product_category_name, product_category_name_english, product_name_length, product_description_length, product_photos_qty, product_weight_g, product_length_cm, product_height_cm, product_width_cm	Stores product information, categories, and physical characteristics
dim_seller	seller_key	seller_id	seller_zip_code_prefix, seller_city, seller_state	Stores seller information and geographic details
dim_payment	payment_key	payment_type	payment_type, payment_installments	Stores payment methods and installment information
dim_order	order_key	order_id	order_status, order_purchase_timestamp, order_approved_at, order_delivered_carrier_date, order_delivered_customer_date, order_estimated_delivery_date	Stores order information and order lifecycle details
dim_date	date_key	full_date	day, day_name, month,	Provides

Dimension Table	Primary Key	Business Key	Main Attributes	Purpose
			month_name, quarter, year, week_of_year	calendar information for time-based analysis

5. Surrogate Keys

Each dimension table contains a surrogate key used to establish relationships with the fact table.

dim_customer

|

| customer_key



fact_sales

dim_product — product_key —► fact_sales

dim_seller — seller_key —► fact_sales

dim_payment — payment_key —► fact_sales

dim_order — order_key —► fact_sales

dim_date — date_key —▶ fact_sales

5. ETL Process

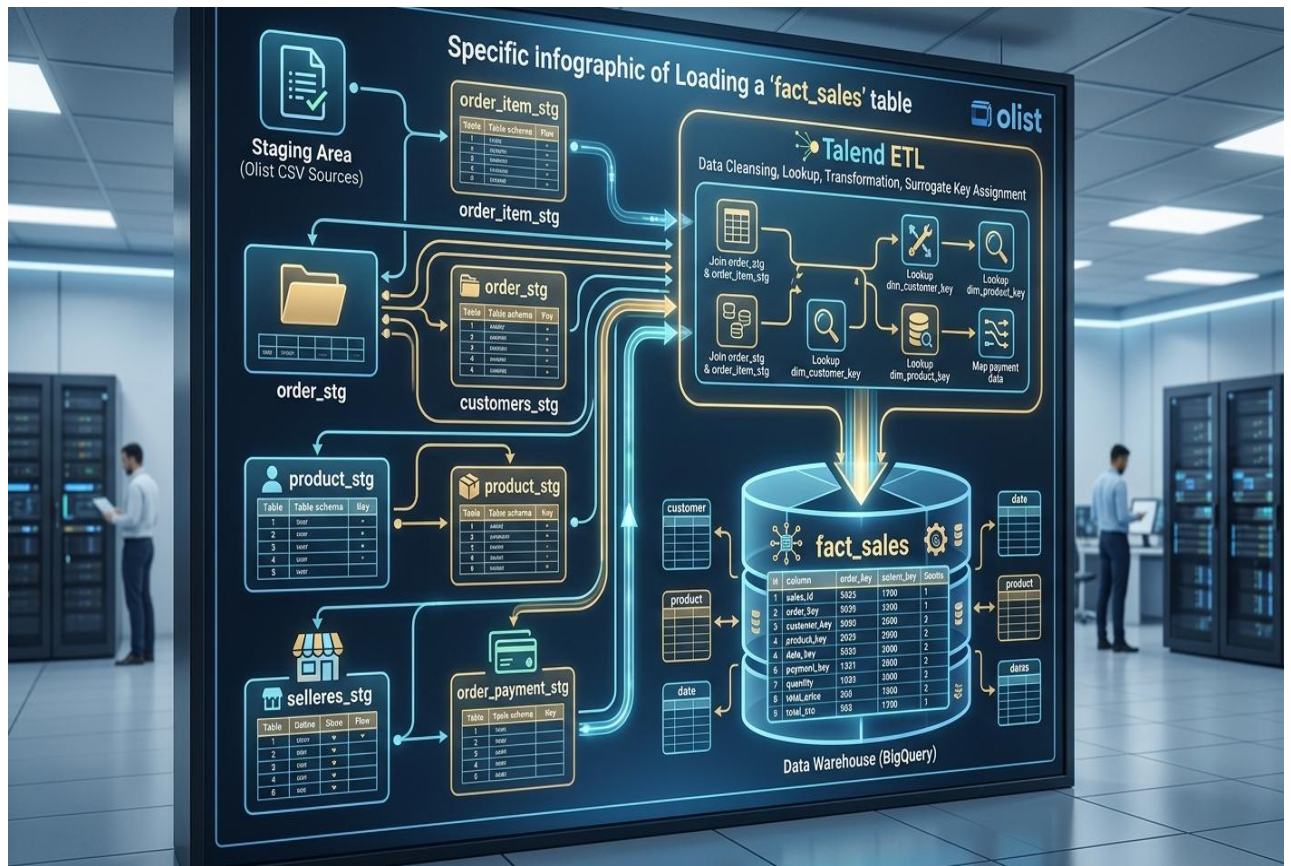


Main Talend components

- **tDBInput**
- **tMap**
- **tDBOutput**
- **tUniqRow**

7. Fact Table Loading

Explain the process you actually implemented:



8. Business Measures

Measure	Description
price	Product price
freight_value	Freight/shipping cost
payment_value	Payment amount
total_item_value	Total item value

```
total_item_value = price + freight_value
```

Data Quality

9. Data Quality & Validation

This section is particularly valuable because you actually encountered and solved these issues.

Include:

During the ETL implementation, several data quality and technical issues were identified and resolved.

Issues addressed

- Float and BigDecimal type conversions
- Long and Integer conversions
- NULL payment values
- Duplicate records
- Duplicate dates in `dim_date`
- Surrogate keys incorrectly loaded as 0
- Date-to-date_key lookup issues
- Dimension-to-fact relationships
- Data type compatibility between Talend and PostgreSQL

This is much better for a portfolio than pretending everything worked perfectly from the beginning.

Final Data Warehouse

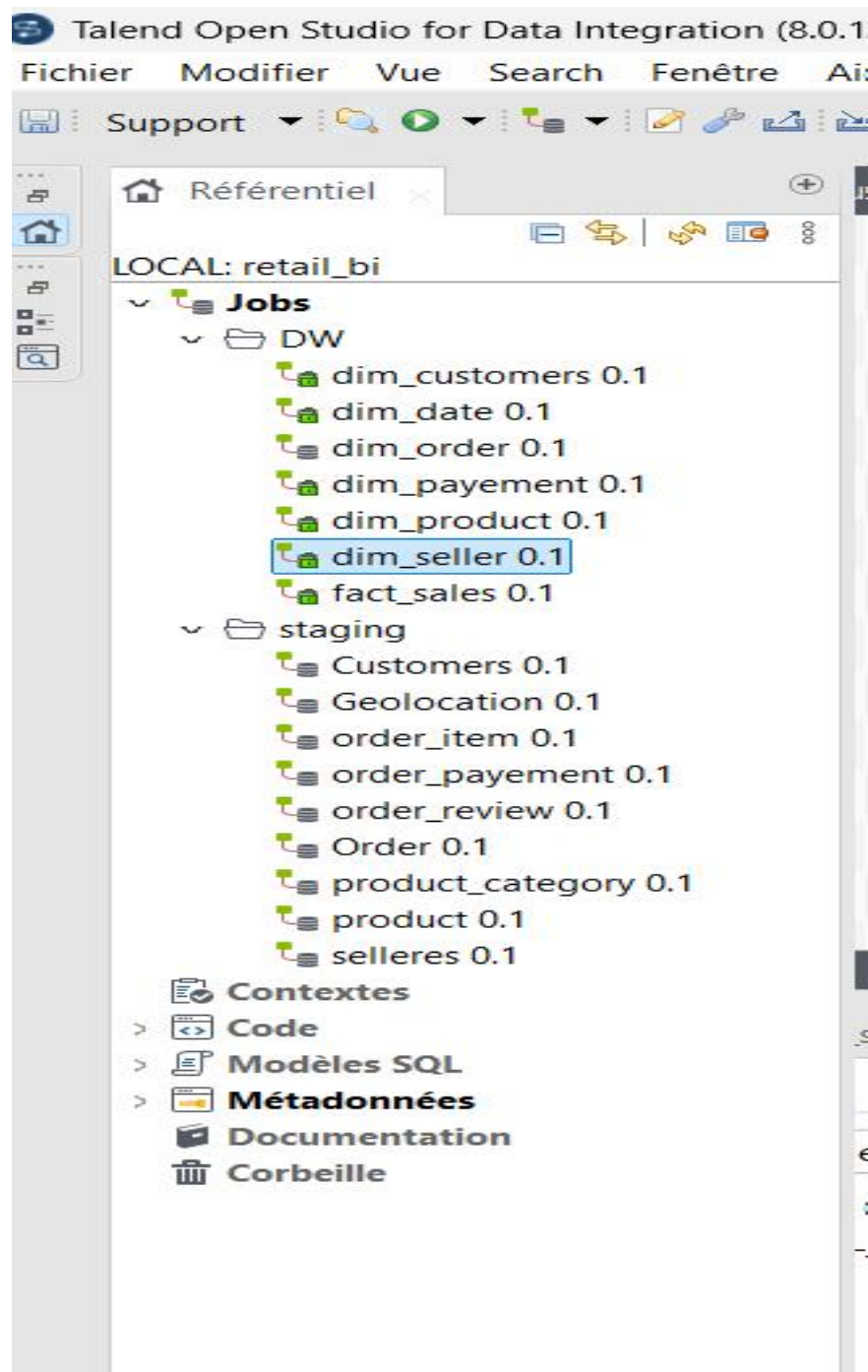
10. Final Result

The final Data Warehouse provides a centralized analytical structure for the Olist e-commerce dataset.

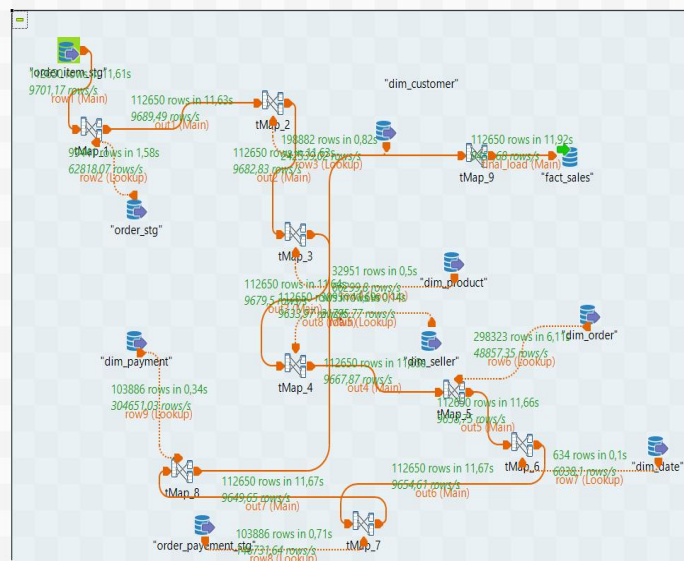
The Star Schema enables efficient analysis of sales according to multiple business dimensions, including customers, products, sellers, payments, orders, and time.

11. Talend_Jobs

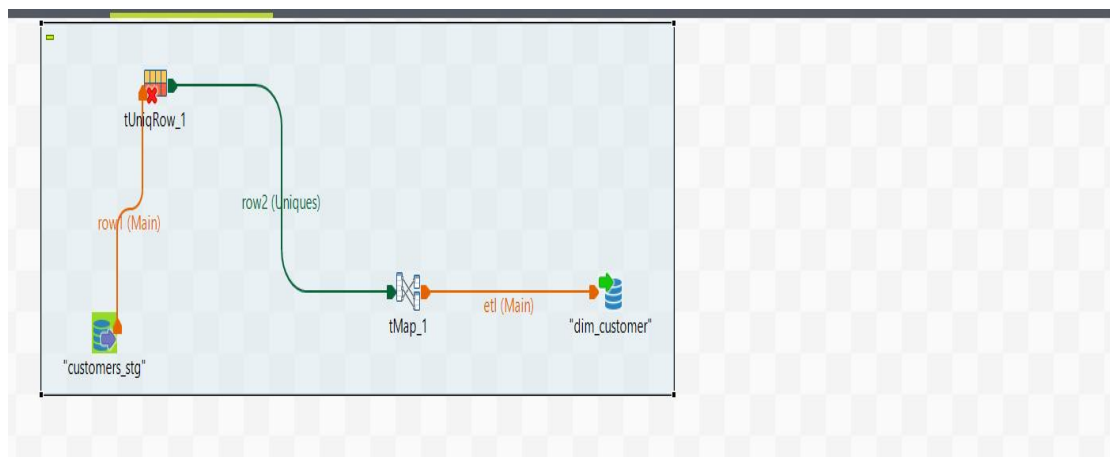
Talend_Architecture_Jobs



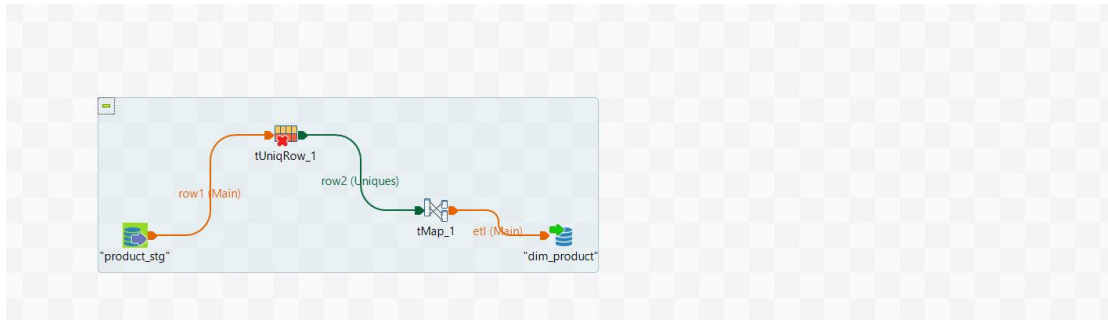
Fact_job



Exemple :dimention_Customers

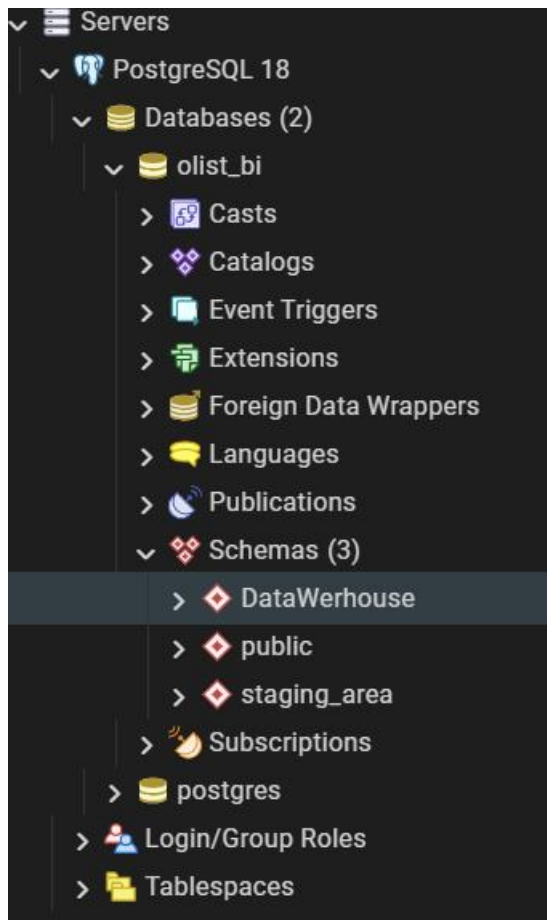


Exemple :dimention_Product



12. PostgreSQL

Achitecture_database



Fact_table

The screenshot shows the pgAdmin 4 interface with the 'DataWarehouse' schema selected in the Object Explorer. The 'fact_sales' table is highlighted under the 'Tables (7)' category. The main query window displays the following SQL query:

```
SELECT * FROM "DataWarehouse".fact_sales
```

The 'Data Output' pane shows the first 8 rows of the table. The table has 12 columns: order_key, customer_key, product_key, seller_key, payment_key, date_key, order_item_id, shipping_limit_date, price, freight_value, and payment_value. The total number of rows is 112650.

	order_key bigint	customer_key bigint	product_key bigint	seller_key bigint	payment_key bigint	date_key bigint	order_item_id integer	shipping_limit_date timestamp without time zone (6)	price numeric (12,2)	freight_value numeric (12,2)	payment_value numeric (12,2)
1	284155	164999	25867	3609	166922	2149	1	2017-09-19 09:45:35	58.90	13.29	
2	270750	133707	27232	3567	166902	2211	1	2017-05-03 11:05:13	239.90	19.93	
3	205186	134397	22625	4920	166924	2308	1	2018-01-18 14:48:30	199.00	17.87	
4	221447	151205	15404	5119	166922	2087	1	2018-08-15 10:10:18	12.99	12.79	
5	204135	107044	8863	4693	166902	2597	1	2017-02-13 13:57:51	199.90	18.14	
6	268349	122439	3940	3755	166925	2218	1	2017-05-23 03:55:27	21.90	12.69	
7	297449	153928	22293	6069	166923	2428	1	2017-12-14 12:10:31	19.90	11.85	
8	246227	117362	6975	3787	166915	2391	1	2018-07-10 12:30:45	810.00	70.75	

Dim_Tables

The screenshot shows the pgAdmin 4 interface with the 'DataWarehouse' schema selected in the Object Explorer. The 'dim_customer' table is highlighted under the 'Tables (7)' category. The main query window displays the following SQL query:

```
SELECT * FROM "DataWarehouse".dim_customer  
ORDER BY customer_key ASC
```

The 'Data Output' pane shows the first 9 rows of the table. The table has 5 columns: customer_key, customer_id, customer_unique_id, customer_zip_code_prefix, and customer_city. The total number of rows is 198882.

	customer_key [PK] bigint	customer_id character varying (50)	customer_unique_id character varying (50)	customer_zip_code_prefix integer	customer_city character varying (100)
1	0668999e2fba1a1fbc88172c00ba8b...	861eef4711a542e4b93843c5dd7fe...	14409	franca	SP
2	18955e8d3d37fde6b2def6b18a428ac...	290c77bc529b7ac935b93aa66c33...	9790	sao bernardo do campo	SP
3	4e7b3e0028585eb08712fdd0374a...	060e732b5b29e8181a18229c7b0b...	1151	sao paulo	SP
4	b2b6027bc5c5109e52944dc6358b1...	259dac757896cd24d7702b9acbf3f...	8775	mogi das cruzeiras	SP
5	4f2d8ab171c80ec83647c12e35b23...	345ecd01c38d18a9036ed96c73b8...	13056	campinas	SP
6	879864dab90c3047522e92c82e121...	4c93744516667ad3b8f1f645a311...			
7	f8b26e7cf63160e36e0908c76c3f441	addec96d2e059c80c30fe6871d30...			
8	5e274e7a0c3809e14aba7ad5aae0d...	57b2a98a409812fe9618067b6b8e...	35182	timoteo	MG
9	5ad08e34b2e993982a47070956c5c...	1175e95fb47ddf9de6b2b0618877...	81560	curitiba	PR

pgAdmin 4

Object Explorer

- Languages
- Publications
- Schemas (3)
 - DataWarehouse
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
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 - Procedures
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 - dim_order
 - dim_payment
 - dim_product
 - dim_seller
 - fact_sales
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 - Types
 - Views
 - public

Query

```
1 SELECT * FROM "DataWarehouse".dim_date
```

Data Output

data_key	full_date	day	day_name	month	month_name	quarter	year	week_of_year
bigint	date	integer	character varying (20)	integer	character varying (20)	integer	integer	integer
1	2085	2017-10-...	2	lundi	10	octobre	4	2017
2	2086	2018-07-...	24	mardi	7	juillet	3	2018
3	2087	2018-08-...	8	mercredi	8	août	3	2018
4	2088	2017-11-...	18	samedi	11	novembre	4	2017
5	2089	2018-02-...	13	mardi	2	février	1	2018
6	2090	2017-07-...	9	dimanche	7	juillet	3	2017
7	2091	2017-04-...	11	mardi	4	avril	2	2017
8	2092	2017-05-...	16	dimanche	5	mai	2	2017
9	2093	2017-01-...	23	lundi	1	janvier	1	2017

Total rows: 634 Query complete 00:00:00.595

Successfully run. Total query runtime: 595 msec. 634 rows affected.

PostgreSQL 18/olist_bi - Database connected

pgAdmin 4

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Query

```
1 SELECT * FROM "DataWarehouse".dim_order
```

Data Output

order_key	order_id	order_status	order_purchase_timestamp	order_approved_at	order_delivered_carrier_date	order_delivered_cus
bigint	character varying (50)	character varying (30)	timestamp without time zone	timestamp without time zone	timestamp without time zone	timestamp without time zone
1	e481f51cbdc54678b7cc49136f2d6...	delivered	2017-10-02 10:56:33	2017-10-02 11:07:15	2017-10-04 19:55:00	2017-10-10 21:25:31
2	53c0b2fcbcb7d0eb06741e2150273...	delivered	2018-07-24 20:41:37	2018-07-26 03:24:27	2018-07-26 14:31:00	2018-08-07 15:27:4
3	47770eb9100c2d0c44946d9cf07ec...	delivered	2018-08-08 08:38:49	2018-08-08 08:55:23	2018-08-08 13:50:00	2018-08-17 18:06:2
4	949d5b44dbf5de918f9c1697b45f...	delivered	2017-11-18 19:28:06	2017-11-18 19:45:59	2017-11-22 13:39:59	2017-12-02 00:28:4
5	ad21c59c0840dec83a9ceb5573f8...	delivered	2018-02-13 21:18:39	2018-02-13 22:20:29	2018-02-14 19:46:34	2018-02-16 18:17:0
6	a4591c265e18cb1d0e452889c2d8a...	delivered	2017-07-09 21:57:07	2017-07-09 22:13:15	2017-07-09 22:13:15	2017-07-09 22:13:15
7	136cc07faa42db2ced53f4c79a6d...	invoiced	2017-04-11 12:32			
8	6514b8ad8028c9f2cc2374de2457...	delivered	2017-05-16 13:10:30	2017-05-16 13:22:11		

Total rows: 298323 Query complete 00:00:02.149

Successfully run. Total query runtime: 2 secs 121 msec. 298323 rows affected.

PostgreSQL 18/olist_bi - Database connected

pgAdmin 4

Object Explorer

- Languages
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Query

```
1 SELECT * FROM "DataWarehouse".dim_payment
```

Data Output

payment_key	payment_type	payment_installments
bigint	character varying (30)	integer
1	63040 credit_card	8
2	63041 credit_card	1
3	63042 credit_card	1
4	63043 credit_card	8
5	63044 credit_card	2
6	63045 credit_card	2
7	63046 credit_card	1
8	63047 credit_card	3
9	63048 credit_card	6

Total rows: 103886 Query complete 00:00:00.522

Successfully run. Total query runtime: 522 msec. 103886 rows affected.

PostgreSQL 18/olist_bi - Database connected

pgAdmin 4

Object Explorer

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Query History

```
1 SELECT * FROM "DataWarehouse".dim_product
2 ORDER BY product_key ASC
```

Data Output

product_key [PK] integer	product_id character varying (50)	product_category_name character varying (100)	product_name_length integer	product_description_length integer	product_photos_qty integer	product_weight numeric (10, 2)
1	1e9e8ef04dbcf1f451ed26657ee517e5	perfumaria	40	287	1	
2	3aa071139cb16b67c9e5dea641aa...	artes	44	276	1	
3	96bd76ec8810374ed1b65e2919757...	esporte_lazer	46	250	1	
4	cef67bcfe19066a932b7673e239eb2...	bebes	27	261	1	
5	9dc1a7de274444849c219c1f195d0b...	utilidades_domesticas	37	402	4	
6	41d3672d4792049fa1779bb35283e...	instrumentos_musicais	60	745	1	
7	732bd381ad09e530fe0a5f457d81be...	cool_stuff	56	1272	4	1
8	2548a7e6e77a690cf3eb6368e9ab6...	moveis_decoracao	56	184	2	

Total rows: 32951 Query complete 00:00:00.555 CRLF Ln 1, Col 1

pgAdmin 4

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Query History

```
1 SELECT * FROM "DataWarehouse".dim_sales
2 ORDER BY seller_key ASC
```

Data Output

seller_key [PK] integer	seller_id character varying (50)	seller_zip_code_prefix integer	seller_city character varying (100)	seller_state character varying (10)
1	3096 344218959a84dea7ee1970632cb2df...	13023	campinas	SP
2	3097 d1b65f7debc3351ea86b5f14c68d2...	13844	mogi guacu	SP
3	3098 ce3ad9de960102d0677a81f5d0bb7b...	20031	rio de janeiro	RJ
4	3099 c0f3ee2e14555b6faee3d58c1b1...	4195	sao paulo	SP
5	3100 51a04a8a6bdc23deccc82b0b8074...	12914	braganca paulista	SP
6	3101 c240c40617171ac1806a6ee72be35...	20920	rio de janeiro	RJ
7	3102 e49c26c3edfa46d227d5121a6b5e4d...	55325	brejao	SP
8	3103 1b938a7acc5a5061a66a37640e75f...	16304	penapolis	SP
9	3104 768a86e36a6aaw3802w3c5433d6...	1529	sao paulo	SP

Total rows: 3095 Query complete 00:00:00.431 CRLF Ln 1, Col 1

Successfully run. Total query runtime: 431 msec. 3095 rows affected.

PostgreSQL 18/olist_bi - Database connected

Technologies

13. Technologies Used

PostgreSQL

Talend Open Studio

SQL

ETL

Dimensional Modeling

Star Schema

Git

GitHub

Conclusion

14. Conclusion

The Data Warehouse phase successfully transformed the prepared Olist data into a dimensional analytical model based on a Star Schema.

The resulting Data Warehouse provides a structured foundation for Business Intelligence reporting and advanced analytics.

The next phase will focus on developing the Business Intelligence layer using Power BI.